

ESP Zigbee Gateway

This tutorial introduces how to use the Module Gateway H2 together with the CoreS3 controller to run the ESP Zigbee Gateway example program. The ESP Zigbee Gateway is a gateway device based on the ESP32 series Wi-Fi SoC and the ESP32-H2 802.15.4 SoC. It can connect Zigbee networks with Wi-Fi networks to enable interoperability among smart home devices.

1. Preparations

- 1. **Environment Setup:** Refer to the [ESP-IDF - Getting Started with ESP32S3 Tutorial](#) to complete the basic compilation environment.

ESP-IDF version

It is recommended to use ESP-IDF version **v5.3.1** for compiling this example.

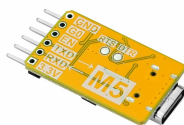
```
git clone -b v5.3.1 --recursive https://github.com/espressif/esp-idf.git
cd esp-idf
./install.sh
. ./export.sh
```

- 2. Use the Git command `clone --recursive` to recursively clone the ESP-Zigbee-SDK repository

```
git clone --recursive https://github.com/espressif/esp-zigbee-sdk.git
cd esp-zigbee-sdk
```

- 3. **Hardware Products Used:**

- [ESP32 Downloader](#)
- [Module Gateway H2](#)
- [CoreS3](#)



- 4. The subsequent tutorial uses `idf.py` commands which depend on ESP-IDF. Before running the commands, execute `./export.sh` in ESP-IDF to activate the related environment variables. For detailed instructions, please refer to the [ESP-IDF - Getting Started with ESP32S3 Tutorial](#).

2. Compiling the RCP Firmware

- 1. Before compiling the Gateway firmware, you need to generate the RCP firmware. Follow the commands below to navigate to the corresponding RCP firmware directory and set the build target to `esp32h2`.

```
cd $IDF_PATH/examples/openthread/ot_rcp
idf.py set-target esp32h2
idf.py menuconfig
```

- 2. Use `idf.py menuconfig` to enter the configuration menu. In menuconfig, configure:
Component config -> OpenThread RCP Example - **Enable OPENTHREAD_NCP_VENDOR_HOOK**
- 3. After completing the configuration, execute the following command to compile the RCP firmware.

```
idf.py build
```

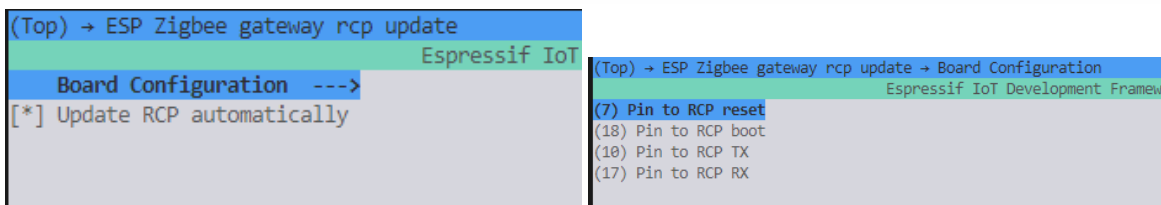
3. Compiling the Gateway Firmware

- 1. Navigate to the `esp_zigbee_gateway` example program directory and set the build target.

```
cd esp-zigbee-sdk/examples/esp_zigbee_gateway
idf.py set-target esp32s3
idf.py menuconfig
```

- 2. In menuconfig, enable the option:
ESP Zigbee gateway rcp update -> Update RCP automatically. Also configure the correct communication pins. This pin configuration is for the CoreS3 controller; if using another controller, modify accordingly.

- Board Configuration
 - Pin to RCP reset: 7
 - Pin to RCP boot: 18
 - Pin to RCP TX: 10
 - Pin to RCP RX: 17



- 3. Gateway Wi-Fi Connection Configuration

- Example Connection Configuration
 - WiFi SSID
 - WiFi Password

```
(Top) → Example Connection Configuration
[*] connect using WiFi interface
[ ] Get ssid and password from stdin
[*] Provide wifi connect commands
(xxxxxx) WiFi SSID
(xxxxxx) WiFi Password
(6) Maximum retry
    WiFi Scan Method (All Channel) --->
    WiFi Scan threshold --->
    WiFi Connect AP Sort Method (Signal) --->
[ ] connect using Ethernet interface
[ ] connect using Point to Point interface
[*] Obtain IPv6 address
    Preferred IPv6 Type (Local Link Address) --->
```

4. Compiling and Flashing

```
idf.py build
idf.py erase_flash
idf.py flash
```

5. Getting Started

- Connect the CoreS3 and the Module Gateway H2.
- Connect the CoreS3 to your computer.
- Use `idf.py monitor` or other serial debugging tools at 115200 bps to view the runtime logs.

```
I (3551) wifi:set rx beacon pti, rx_bcn_pti: 0, bcn_timeout: 25000, mt_pti: 0, mt_time: 10000
I (3561) wifi:AP's beacon interval = 163840 us, DTIM period = 1
I (3561) wifi:<ba-add>idx:0 (ifx:0, c4:5e:5c:9b:63:f0), tid:6, ssn:2, winSize:64
I (3611) wifi:<ba-add>idx:1 (ifx:0, c4:5e:5c:9b:63:f0), tid:0, ssn:0, winSize:64
I (4651) esp_netif_handlers: example_netif_sta ip: 192.168.52.21, mask: 255.255.255.0, gw: 192.168.52.1
I (4651) example_connect: Got IPv4 event: Interface "example_netif_sta" address: 192.168.52.21
I (5461) example_connect: Got IPv6 event: Interface "example_netif_sta" address: fe80:0000:0000:0000:4a27:e2ff:fe78:fd74, type: ESP_IP6_ADDR_IS_LINK_LOCAL
I (5461) example_common: Connected to example_netif_sta
I (5461) example_common: - IPv4 address: 192.168.52.21,
I (5471) example_common: - IPv6 address: fe80:0000:0000:0000:4a27:e2ff:fe78:fd74, type: ESP_IP6_ADDR_IS_LINK_LOCAL
I (5481) wifi:Set ps type: 0, coexist: 0

I (5481) ESP_ZB_GATEWAY: Initialize Zigbee stack
W (5511) ESP_ZB_GATEWAY: Network(0xb3f7) closed, devices joining not allowed.
I (5511) ESP_ZB_GATEWAY: Device started up in non factory-reset mode
I (5511) ESP_ZB_GATEWAY: Device rebooted
I (6121) ESP_ZB_GATEWAY: Network(0xb3f7) is open for 180 seconds
```

Normal runtime log contents:

- RCP firmware version check
- Successful Wi-Fi connection
- Successful Zigbee network creation
- Network open to allow device joining